Final Assignment: Unsupervised Machine Learning Report

Main Objective: The objective of this analysis is to segment customers based on purchasing behavior using clustering techniques. This helps businesses target customers effectively and improve marketing strategies.

Dataset Description: The dataset contains customer information such as age, annual income, and spending score. The goal is to identify distinct customer groups.

Data Exploration & Preparation: Missing values were handled, data was normalized, and features like income and spending score were selected. Outliers were analyzed.

Model Variations:

1. K-Means Clustering: Tested different values of K using Elbow Method.

2. Hierarchical Clustering: Used dendrogram to identify clusters.

3. DBSCAN: Identified density-based clusters and noise.

Final Model Recommendation: K-Means performed best due to clear cluster separation and simplicity.

Key Findings & Insights:

- Customers can be divided into high-value and low-value groups.

- High-income high-spending customers are most profitable.

- Low-income low-spending customers need engagement strategies.

Limitations: Limited features and possible noise in data.

Next Steps:

- Add more features like purchase history.

- Try PCA for dimensionality reduction.

- Deploy model for real-time segmentation.